Inventoria User Manual

1. Requirements

- Windows 10/11
- Python 3.10 or later, with Tkinter
- XAMPP with its MySQL/MariaDB service
- mysql-connector-python (installed from requirements.txt)

2. Installation and first run

1. Install Python and XAMPP. In the XAMPP Control Panel, start MySQL. Apache is only needed to use phpMyAdmin.
2. Open PowerShell in the project folder and create the virtual environment:

```
powershell
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
```

3. Configure the MySQL host, port, username, and password in config.py if they differ from the XAMPP defaults.
4. Create the database and first administrator:

```
powershell
python setup_database.py
```

The setup prompts for an administrator username, full name, and password. Password input is hidden, the password is never printed or written to a file, and it must be 12-128 characters. If importing database/inventoria_schema.sql through phpMyAdmin, run this setup command afterward to create the first administrator.

5. Launch the application:

```
powershell
python main.py
```

Or use run_inventoria.bat after creating the virtual environment.

3. Sign-in and roles

Enter the administrator credentials created during setup. Staff can create an account from the login screen; self-registered accounts receive the Staff role. Administrators can manage accounts, assign roles, and edit role permissions from Users.

The navigation and controller permission checks restrict administrative actions. Staff may view/search products, record stock movements, and view their own transactions. Administrators can manage products, users, reports, audit logs, and all transactions.

4. Common tasks

Find products

Open Products, type a product name, specification, SKU, or category in the search field, then select a row to edit it if permitted.

Add or update a product

Administrators select + Add Product or Edit. Enter the product name, selling-unit specification, SKU (optional), category, price, and low-stock threshold. Opening stock is recorded as a stock-in transaction. Product stock is intentionally not editable in the product form.

Record stock in/out

Open Stock, choose a product and movement type, enter a positive whole-number quantity, then submit. Stock-out cannot exceed the available quantity. Each movement updates stock, creates a transaction, and writes an audit entry in one database transaction.

Search and export transactions

Open Transaction Records, choose filters, and select Search. Administrators can filter by user; Staff only see their own records. Use Export CSV to save the visible filtered results.

Generate and export a report

Administrators open Reports, choose a date range, and select Generate. The report shows stock totals, movement totals, stock value, and low-stock status. Select Export CSV to save it.

Manage users and audit history

Administrators use Users to add accounts, change roles, activate/deactivate accounts, reset passwords, change permissions, or create/restore database backups. Use Audit Log to review recorded changes.

5. Tests

Run the unit tests from the project folder with:

```
python -m unittest discover -s tests -v
```

The tests cover active domain objects, repository success and failure paths, password handling, validators, and legacy service logic. MySQL and a graphical display are not required for the unit suite.

6. Data, logs, and security

• MySQL/MariaDB is the primary store. The SQL schema is in database/inventoria_schema.sql.
• Passwords are stored as salted PBKDF2-HMAC-SHA256 hashes.
• Application diagnostics are written to inventoria.log, rotated at 1 MB with five backups.
• Audit events are stored separately in MySQL.
• Backup JSON includes password hashes and business records; keep it private.

7. Troubleshooting

• Cannot reach MySQL: start MySQL in XAMPP and verify DB_CONFIG in config.py.
• Missing database or tables: run python setup_database.py.
• No administrator exists: run python setup_database.py and complete the hidden-password prompt.
• Access denied: ask an administrator to review your account role and permissions.
• Unexpected application error: review inventoria.log; the UI also reports the error.